Unnamed_Unit

Unit 1 Conditional Statements "If / If Else / Nested If"

PDF Documents

PDF Document 1: 1731842025804-Chapter 3_ unit1 Conditional Statements.pdf

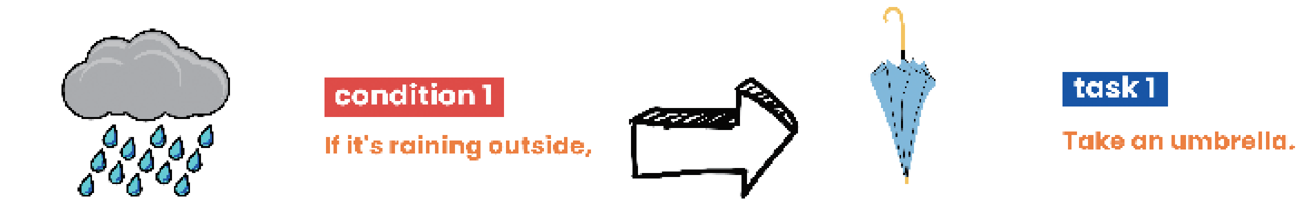
This PDF document is included in the unit materials.

UNIT 1 CONDITIONAL STATEMENTS

If else Statement

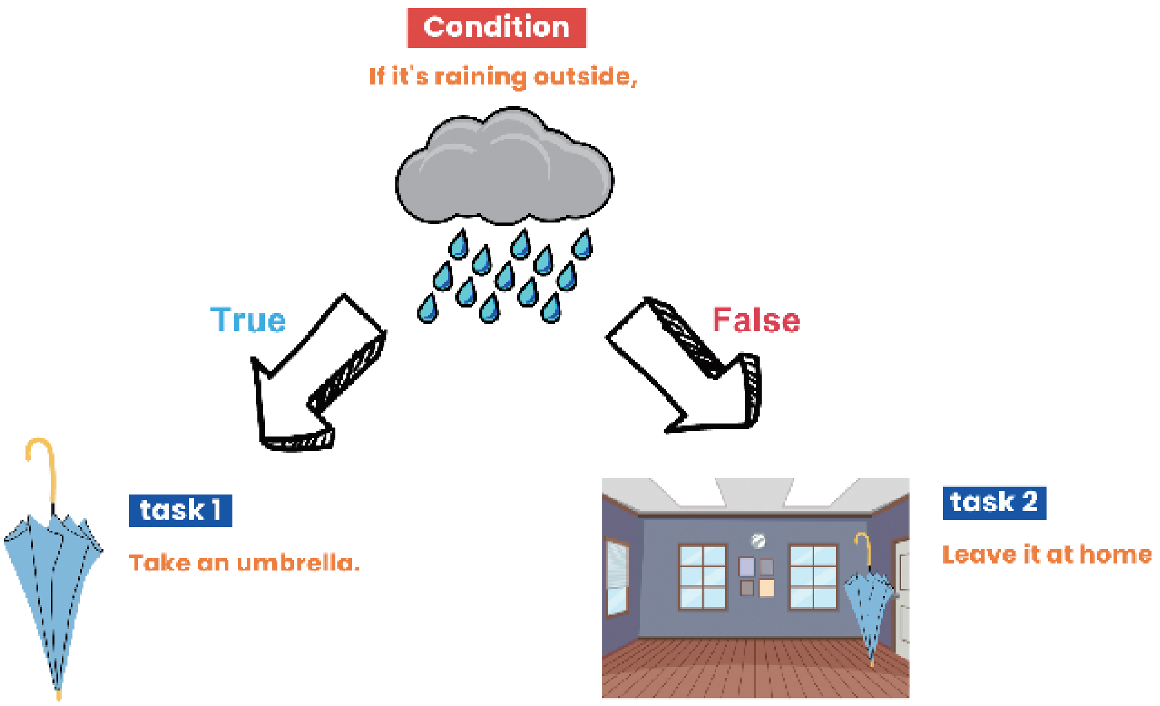
An 'If-Else' statement is like an 'If' statement with a backup plan. If the 'If' part isn't true, then the 'Else' part takes over and does something else instead.

✓ ELSE



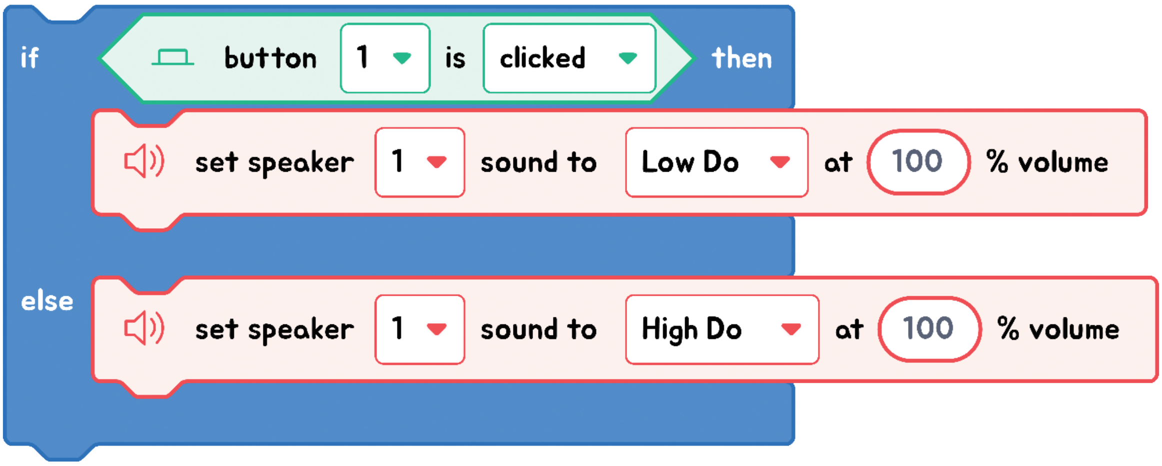
Imagine you're deciding whether to bring an umbrella. An 'If-Else' statement can help you with this!

If it's raining outside (this is the condition), you'll take an umbrella (this is the task). Otherwise (this is the Else part), you'll leave the umbrella at home. So, even though we only consider rain, Else takes care of everything else by default.

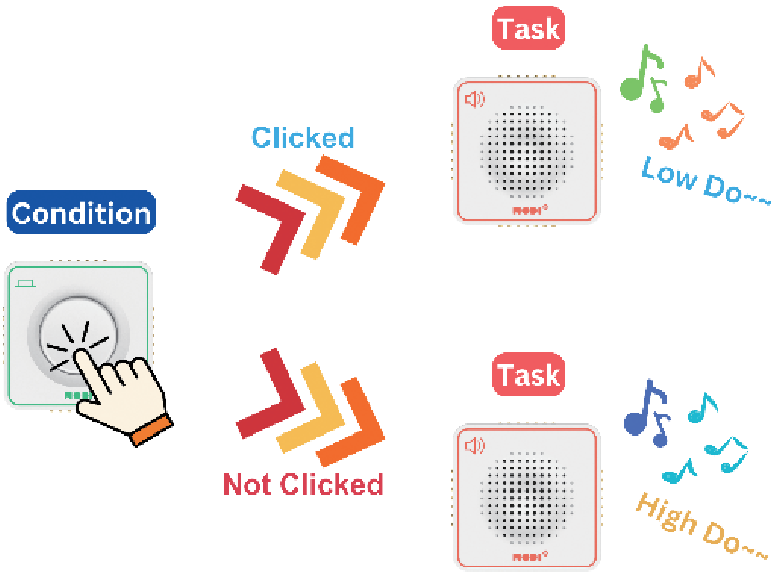


The key to 'If-Else' is that if the 'If' condition isn't met, the program will do what's in the 'Else' part instead.

EXPLANATION WITH CODE EXAMPLE



In block coding, the "If-Else" statement lets us choose between two actions based on a condition. If the condition is true, one action is taken; if not, a different action is executed. The "If-Else" block visually splits this choice, ensuring the program always has a clear direction, no matter the condition's outcome.

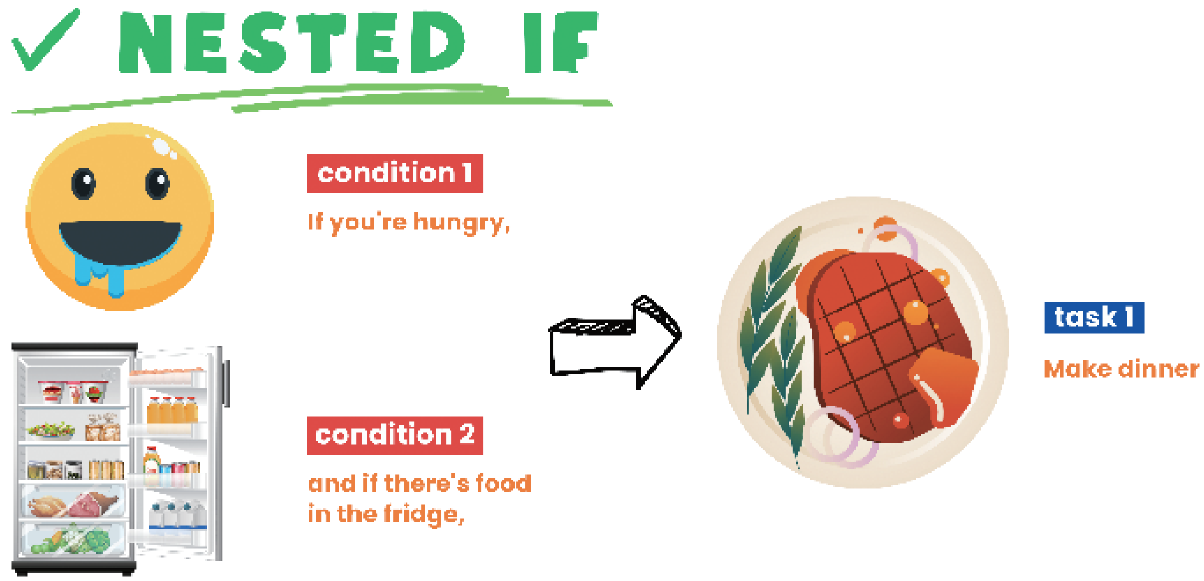


Code Insight

1. **Condition Checking:** The program starts by checking a specific condition, like seeing if a button has been pressed. So, the condition in the code is "If the button is clicked."
2. **Execution if True:** If the condition is met, then the instructions inside the "if" statement are carried out. These instructions specify the actions to take when the condition occurs. Thus, the task performed is "Set the speaker to play the 'low do' sound at 100% volume."
3. **Execution if False:** If the condition is not met, then the instructions inside the "else" statement are carried out. These instructions specify the actions to take when the condition is not met. Thus, the task performed is "Set the speaker to play the 'high do' sound at 100% volume."

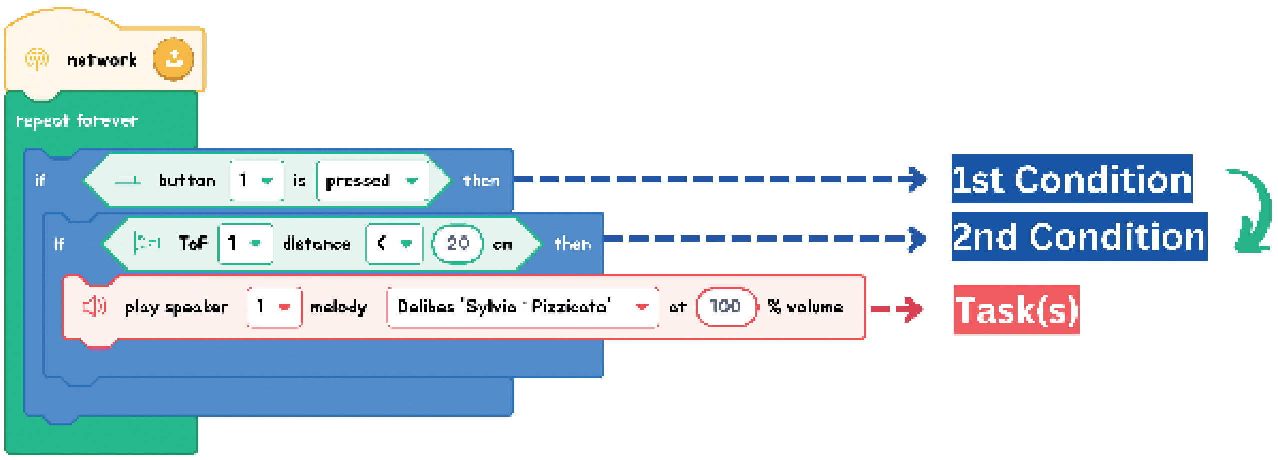
Nested If Statement

Nested if statements are like putting one decision inside another. First, you check a condition; if that's true, you check another one inside it. If this second condition is also true, then an action takes place. It's like solving a layered puzzle: solve one layer to get to the next!



First, you check if you're hungry (1st condition). If you are, you then look inside the fridge to see if there's food (2nd condition). If there's food available, you go ahead and make dinner (this is the task).

So, with a 'Nested If', you first decide if you're hungry, and only then do you consider whether there's food in the fridge to make a meal. Let's look at the example below!

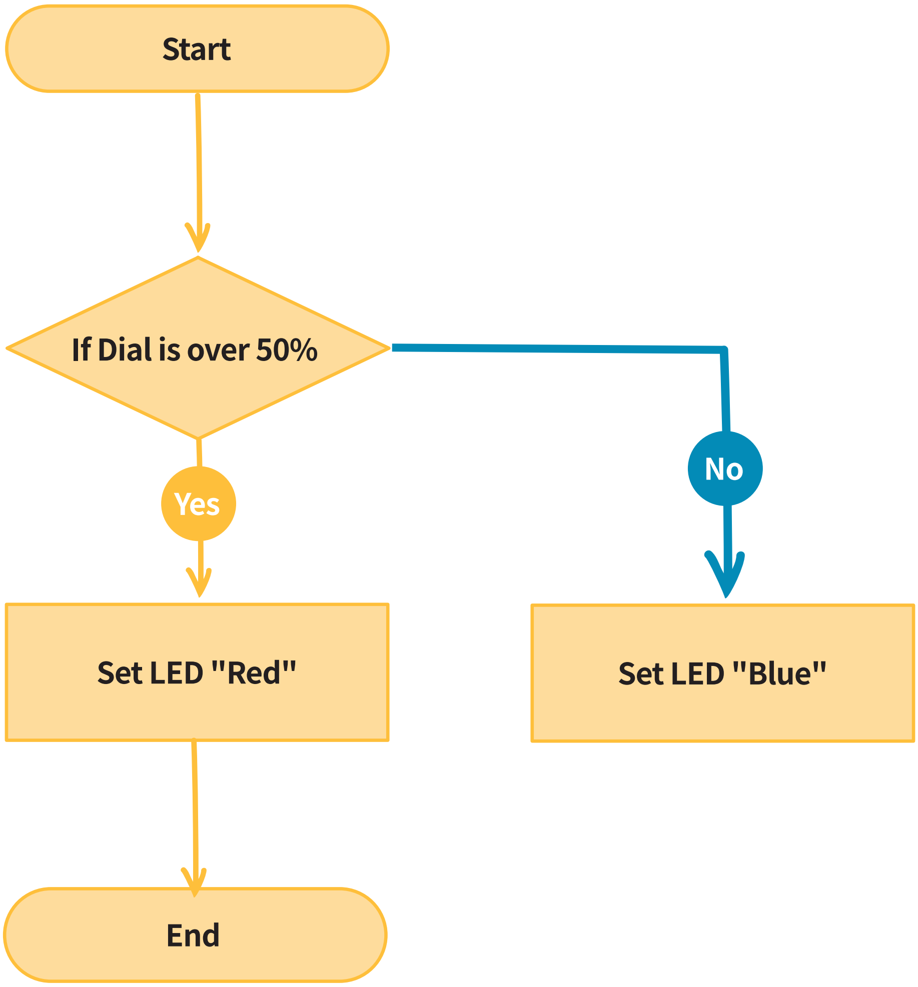


First, the program checks if a button is pressed (1st condition). If the button is indeed pressed, it then examines another condition: whether the distance measured by the ToF sensor is less than 20 cm (2nd condition). Only if both these conditions are met, does the program command the speaker to play the song.

Exercise

Exercise 1

Find the condition and task in the flowchart.



Your Answer:

Condition(s)

Task(s)